OAL Trinity Currency Laboratory — Independent Question-to-Experiment and Falsification Directive

2026-11-08

OAL Trinity Currency Laboratory

Independent Question-to-Experiment, Assumption-Challenge, and Experimental Falsification Directive

Document ID: OAL-CURR-RESEARCH-INVESTIGATION-001
Version: 1.0
Delivery edition: REGENERATED-02 (second replacement; November 8, 2026)
Issued: November 8, 2026
Status: Pre-Canon Research and Experimental Directive
Prepared by: ChatGPT (OpenAI GPT-6), at the express request of the OAL Trinity Currency Laboratory Project Owner
Participants: ChatGPT, Claude AI, Google Gemini, Microsoft Copilot, Perplexity AI, and Grok AI
Intended publication: OAL-AI-UNIFIED-ARCHIVE / Trinity Currency Laboratory project folder

> **Publication and interpretation notice.** This document establishes a method for inquiry. It does not adopt a final currency architecture, certify extraordinary physical or financial capabilities, establish enforceable financial rights, or authorize real-world financial transactions. Results produced under it are research artifacts, not automatically project canon.

Executive Summary

The OAL Trinity Currency Laboratory investigates a proposed *Quantum Credit* (QC), a target-relative value concept, alongside proposed roles for *1COIN* and *NCOIN*. Earlier work produced architectural drafts, model assumptions, test pilots, verifier code, audit findings, and question registers asking the Project Owner to select among competing choices. Reviewing these questions exposed a methodological problem: many matters described as owner decisions are better treated first as experimental questions.

This directive moves the laboratory from **decision-first implementation** toward **investigation-first evaluation**. Each AI collaborator is asked to revisit its own earlier questions, examine hidden assumptions, develop independently chosen tests, and explicitly permit failure of its recommended answers, the framing of the question, the underlying mechanism, and the evaluator used to judge it. Contributors are **not** required to adopt a universal run card, identical benchmark, common operating paradigm, or unanimous conclusion. Experiments must document local assumptions, preserve negative and inconclusive findings, and distinguish observed behaviors from stipulated capabilities.

The principal deliverable is not an approved design. It is an evidence record that makes later design and owner decisions better informed.

1. Background and Project Context

1.1 Conceptual origin

The laboratory's working concept is that QC resolves value relative to a specified target, rather than simply functioning as a fixed-denomination conventional token. The concept is frequently summarized as:

QC(T) = V(T)

Here, *T* denotes a target, and *V(T)* a target-relative valuation or resolution concept; this notation is a research premise, **not** a proof of executable value creation, delivery, ownership, or physical possibility. The project also explores a proposed trinity of instruments: QC for target-relative resolution, 1COIN for settlement or completion, and NCOIN for negative consequence, countervalue, or unresolved effects. The necessity, independence, and exact boundaries of these instruments remain open to examination.

Prior architecture discussions considered a sequence resembling target resolution, classification, valuation, synchronization, transferability, seller or authority, authorization, resolution, settlement, and consequence. They also explored an *Unpurchasable Domain* and an *Absolute Transaction Boundary*, including targets with no legitimate seller, nontransferable properties, incoherent descriptions, or impossible outcomes. These are research areas rather than universally settled rules.

1.2 Prior research and the decision bottleneck

The collaboration developed multiple proposed baselines and laboratory approaches, including local simulations and question registers prepared for the Project Owner. Those registers raised consequential questions: what QC must accomplish; whether to select one common pilot; who may authorize resource creation; what qualifies as a useful delivery; how interrupted transactions should behave; what constitutes the same request; how QC, 1COIN, and NCOIN relate; and what evidence constitutes settlement.

The Project Owner clarified on November 8, 2026 that answering many of these questions by choosing one of the offered options would be premature. Instead, the laboratory should test the assumptions embedded in the questions and the recommendations, including the possibility that **all** proposed options fail. The Project Owner also expressly declined to select a universal run card or require the collaborators to converge upon one consensus approach.

1.3 Lessons from prior code verification

Previous pilot work reported successful local checks, but follow-on adversarial audit work found that some deliberately altered records could be accepted by checkers. This distinction matters: a passing verifier is evidence about the implementation and conditions **actually checked**; it does not, by itself, establish the correctness of the specification or the intended capability. A checker defect must not be confused with proof that a currency concept is impossible; conversely, a successful simulation must not be represented as proof that an extraordinary capability exists outside the model.

1.4 Historical sources and limits

This directive is derived from the Project Owner's November 8 clarification; the discussion and consolidated directives attributed to Perplexity, Grok, Microsoft Copilot, Google Gemini, and Claude; and recoverable context from the OAL Currency Laboratory work, including the ChatGPT owner-question register identified as OAL-CURR-OWNER-QUESTIONS-001 and associated pilot/audit discussions. The document is an integrated new directive, **not** a claim that every historical repository attachment or every AI's underlying question file was independently reopened and audited in full. Each participant must report its own source-access limitations when executing the directive.

2. Purpose, Research Objectives, and Scope

Purpose. Turn unresolved questions into testable or analytically investigable research tasks before demanding premature owner commitments.

Research objectives. (1) distinguish owner judgments from experimentally investigable questions; (2) expose hidden premises; (3) compare or challenge independently selected mechanisms; (4) test whether evaluators can detect relevant failures; (5) preserve successful, failed, mixed, and inconclusive evidence; and (6) use that evidence to refine later questions and decisions.

Within scope: conceptual analysis; bounded simulations; model checking; deterministic replay; controlled code and checker audits; adversarial input and boundary-case testing; comparative evaluations; and independent documentation of evidence.

Outside scope without separate authorization: live financial transactions, genuine issuance or transfer of monetary instruments, accessing third-party systems without authorization, altering the public project repository, publishing on the owner's behalf, or declaring a simulated extraordinary result to be physically demonstrated.

3. Governing Research Principles

1. **Investigation before owner commitment.** An unanswered technical question is not automatically a blocker. Contributors should first determine what evidence can be developed without a permanent owner choice.
2. **Independent pathways.** Each collaborator may select its own run card, custom configuration, analytical framework, or different subset of questions. Comparability is desirable when meaningful; uniformity is not mandated.
3. **No forced consensus.** Agreement and disagreement are both permissible. Neither majority vote nor contrived disagreement establishes truth.
4. **Recommendations may fail.** Tests cannot have success for the preferred recommendation hard-coded as their only permissible outcome.
5. **Questions may fail.** An investigation may establish that an original question is ill-formed, premature, assumption-dependent, or missing important alternatives.
6. **Evaluators may fail.** A verifier, metric, or benchmark is part of what must be challenged, not an unquestioned source of authority.
7. **Local assumptions stay local.** Provisional rules are not project canon, and freezing one experiment's setup does not require all six investigators to use that setup.
8. **Failure is data.** Preserve negative results and distinguish mechanism defects, assumption defects, checker defects, and uninformative experimental designs.
9. **Claims follow evidence.** Keep intended capability, stipulation, implementation, observed output, interpretation, and supported conclusion separate.
10. **Owner authority remains intact.** Laboratory evidence may inform values, governance, publication, and consequential authorization decisions, but does not substitute for them.

4. Mandatory Self-Audit of Prior Owner Questions

Each contributor shall identify the question document it previously prepared (title, date, version or identifier if known), list the original questions it selects for further study, and reconsider which questions genuinely require the Project Owner.

Use these categories as **descriptors, not necessarily mutually exclusive bins**:

- **Owner-determinable:** value, governance, narrative priority, legitimacy, or consequential authorization.
- **Experimentally informable:** empirical or simulated evidence can improve later owner judgment.
- **Experimentally discriminable:** defined tests can distinguish alternatives under declared criteria.
- **Assumption-dependent:** plausible answers change under unresolved premises.
- **Premature or ill-formed:** the question lacks usable definitions, evidence, or coherent framing.
- **Contributor-resolvable:** a temporary technical decision can be made locally without binding the whole project.
- **Question under challenge:** the wording or its presuppositions are themselves objects of examination.

For each selected question, record: (a) original wording or an explicitly labeled reconstruction; (b) suggested options or defaults; (c) hidden assumptions; (d) excluded alternatives; (e) investigation method; (f) what could disprove or weaken the proposed response; and (g) whether a genuine owner decision is presently needed.

If your source file is unavailable, state this clearly. Do not invent exact quotations, titles, or question numbers.

5. Independent Experimental Design Protocol

5.1 Selection

A collaborator may adopt its prior run card, construct another run card, choose no run card, or use a different experimental abstraction. Briefly justify why the selected configuration is informative. There is no mandatory shared pilot, frozen benchmark, or operating paradigm for this phase.

5.2 Before execution

Record an experiment identifier, investigator, version, research question, hypothesis, alternatives (including *none of the above*), local assumptions, inputs, expected outcomes, evaluation methods, and explicit failure and inconclusive conditions. Use the label **[LOCAL EXPERIMENTAL ASSUMPTION]** for rules that have not been adopted by the Project Owner.

5.3 During execution

Retain test inputs, environment and software versions where applicable, logs or traces, source code where shareable, validator outputs, and discrepancies. Where possible, run negative or mutated controls aimed at causing the evaluator to make a mistake. Do not silently redefine a passing result or alter assumptions after observing outcomes.

5.4 After execution

Differentiate a proposed test, an analytical argument, a simulation actually executed, an observed result, an independently checked result, and a claim remaining untested. Preserve failed original runs and version later revisions separately. If execution is unavailable, provide an actionable test specification and label the work *not executed*.

6. Mandatory Failure and Falsification Architecture

For every major hypothesis, mechanism, recommendation, or evaluation method, explicitly allow and assess:

Failure or uncertainty mode	Laboratory meaning
Mechanism failure	The proposed operation fails to achieve declared behavior under tested conditions.
Recommended-option failure	One proposed answer performs inadequately.
All-options failure	Every supplied option fails; a new alternative or question is required.
Assumption failure	A premise becomes contradictory, unsupported, or inapplicable.
Question-framing failure	The original question is premature, misleading, or depends on an invalid presupposition.
Evaluator failure	A checker reports false success/failure or fails to observe the intended property.
Experimental-design failure	The setup cannot meaningfully distinguish competing outcomes.
Inconclusive result	Evidence does not support a discriminating conclusion.

Also identify evidence that would strengthen the hypothesis, evidence that would require revision, and a threshold or condition for recommending abandonment. Where a question concerns values, investigate consequences and trade-offs without mislabeling a value choice as empirically falsifiable.

7. Research Domains for Independent Investigation

These domains are **illustrative**, not a required shared suite. Contributors may select, subdivide, combine, or replace them with more informative investigations.

7.1 QC intended capability and target resolution

Review documented goals before asking the Project Owner to redefine QC. Test whether proposed target-relative valuations have coherent semantics for ordinary, nontransferable, disputed, and impossible targets. Preserve the difference between stipulating an outcome in software and demonstrating a real-world capability.

7.2 Authority over resource creation

Investigate centralized, distributed, conditional, adaptive, deliberately unrestricted, or alternative authority models. Examine misuse, contradictions, identity, accountability, and whether existing creation assumptions improperly constrain the intended capability. Unspecified authority is not the same as intentionally unrestricted authority.

7.3 Usable and acceptable delivery

Distinguish recorded balances, valid resource descriptions, actual availability, usable delivery, recipient acceptance, and settlement. Deliberately test cases where technical completion does not produce a usable outcome. Permit the conclusion that no investigated criterion is adequate.

7.4 Interrupted or partial delivery

Compare, where meaningful, rollback, resumption, partial finality, compensation, deferred resolution, and alternative representations. No contributor must adopt a common recovery policy. Identify which behaviors are permitted under each local model and when a policy fails.

7.5 Repeated requests and obligation identity

Challenge definitions based on request ID, payload, target, intention, obligation, state change, or causal history. Test retries, duplicates, replay, legitimate repeat demand, altered requests, and ambiguous identity. A convenient coding convention is not automatically a correct conceptual definition.

7.6 QC, 1COIN, and NCOIN necessity

Test whether instruments have independent functional roles, whether combining or removing instruments introduces errors, and whether some roles depend on narrative or governance requirements not observable in a simplified ledger. The survival of all three instruments is not guaranteed.

7.7 Temporal models, recursion, and transaction boundaries

Investigate static, dynamic, commitment-dependent, or relational valuation; sequencing, finality, reversal, feedback, conservation, and recursive consequences. Test boundaries involving no seller, no legitimate authority, nontransferability, incoherence, or impossibility. Do not silently equate conceptual modeling with real-world permissibility.

8. Evaluator and Code-Audit Requirements

Where there is executable or formal evaluation, challenge the evaluator as actively as the system. Appropriate methods include malformed output records, mutated traces, corrupted states, independent expected-value calculations, invariance/metamorphic checks, negative controls, and competing validators.

Every material test report should distinguish at least: **(1) mechanism or conceptual failure; (2) implementation bug; (3) checker/evaluator weakness; (4) underspecified contract; (5) behavior deliberately permitted by a local policy; and (6) limitation of the experiment.** A successful checker run is not conclusive evidence of a successful currency architecture. A checker failure is not proof of impossibility of the broader concept.

9. Evidence Discipline and Inter-Contributor Review

Each contributor must report all material favorable, unfavorable, and inconclusive findings, including what the evidence **does not** establish. Use a stable experiment identifier and version for each local investigation. Avoid changing criteria after results are known without preserving the previous run.

For cross-contributor review, describe a finding another AI can challenge or replicate **without requiring that reviewer to adopt the same overarching architecture**. When comparisons involve different environments, definitions, or metrics, report those differences before comparing conclusions. Do not average incompatible results into a supposed consensus.

10. Project Owner Decisions and Governance

Contributors should proceed with local, reversible, bounded experiments when that is responsible. Escalate a question to the Project Owner only where it requires a genuine value or governance decision, material authorization, or irreducible project-specific information. For each escalated question, explain what evidence could inform it, why experimentation alone cannot determine it, and whether other investigations can continue.

Nothing in this directive permits unauthorised external access, real transactions, real-world asset creation, publication, or repository modification. The Project Owner retains authority over the official project roadmap and whether experimental results become adopted architectural requirements.

11. Required Deliverables and Report Structure

Each AI collaborator should produce an independent, self-contained report covering:

1. **Source verification:** exact prior question document reviewed; records accessible; limitations.
2. **Question self-audit:** reclassification and presupposition analysis.
3. **Question-to-investigation map:** original questions, hypotheses, alternatives, methods.
4. **Independent approach:** selected run card, model, or other method and rationale.
5. **Local assumptions:** labeled and justified experimental stipulations.
6. **Failure architecture:** mechanism, option, assumption, question, evaluator, and inconclusive conditions.
7. **Execution status:** planned versus actually performed experiments.
8. **Adversarial evaluator audit:** attempts to defeat or contradict the checks.
9. **Results and artifacts:** observations, traces, code/evidence where applicable, unfavorable outcomes.
10. **Interpretation and limits:** supported and unsupported conclusions.
11. **Reassessment:** recommendations retained, revised, or abandoned; reframed questions.
12. **Cross-AI comparison hooks:** specific reproducible or challengeable findings.
13. **Residual owner decisions:** only genuinely owner-dependent items.
14. **Recommended next investigation:** selected to reduce the most important remaining uncertainty.

Do not claim to have run tests that were only specified or imagined. If a deliverable file format is unavailable, supply machine-readable or plain-text material with an explicit limitation.

12. Recommended Collaboration Cycle

Stage 1 — Independent self-audit. Each AI inspects its own questions before receiving the new round's analyses from others.

Stage 2 — Independent tests. Each AI executes or specifies investigations under its own declared assumptions.

Stage 3 — Adversarial exchange. Contributors review other investigators' mechanisms, assumptions, evidence, and evaluator validity.

Stage 4 — Comparative evidence review. Identify aligned, conflicting, non-comparable, and unresolved findings without enforcing consensus.

Stage 5 — Owner decision review. Return only the genuine judgments or authorizations that remain, supported by the evidence produced.

This staged cycle is a **recommended coordination workflow**, not a constraint forcing identical experiments.

13. Success Criteria and Closing Directive

Success is **not** defined by demonstrating that QC works, proving it cannot work, retaining every instrument, passing every validator, choosing one run card, or persuading all six AIs to agree. Success is an investigation that makes the state of knowledge more discriminating: a mechanism may fail; an assumption may collapse; all proposed options may be inadequate; a question may need reformulation; or a checker may prove unreliable.

Directive to the team: Revisit your own question register. Replace premature decision requests with independent, inspectable investigations wherever possible. Permit your mechanisms, questions, assumptions, recommendations, and tests to fail. Preserve what happens. Report only what the evidence supports. Do not make unanimous agreement or a single shared laboratory pathway the objective.

A failed mechanism can yield useful data. A failed assumption can improve an architecture. A failed question can reveal a better one. A failed evaluator can prevent a false conclusion.

Appendix A — Suggested Experiment Record Schema

Use fields adapted to the selected method; these are reporting aids, not a universal run card.

Field	Description
Experiment ID / version	Stable local identifier and revision.
Source question	Exact prior question, or marked reconstruction.
Investigator	AI collaborator and relevant tools.
Hypothesis / alternatives	Competing claims, including none-of-the-above if applicable.
Local assumptions	Explicit [LOCAL EXPERIMENTAL ASSUMPTION] entries.
Inputs / environment	Initial state, parameters, code versions, model boundaries.
Predictions / observables	Measurements and expected differences.
Failure provisions	Rejection, revision, inconclusive, checker-failure conditions.
Evaluator challenge	Negative controls, mutations, independent checks.
Observed evidence	Logs, state traces, outputs, derivations, run status.
Interpretation / limits	Narrowest justified conclusion and known limitations.
Follow-on work	Next discriminating experiment.

Appendix B — Source and Attribution Notes

- **Primary owner clarification:** November 8, 2026, request to translate unanswered owner questions into independent experiments designed to permit recommendation, assumption, question, and evaluator failure.
- **Collaborative drafts reviewed:** responses attributed to Perplexity AI, Grok AI, Microsoft Copilot, Google Gemini, and Claude AI within the supplied collaborative text.
- **ChatGPT historical reference:** OAL-CURR-OWNER-QUESTIONS-001, November 7, 2026, and associated Currency Laboratory pilot and verifier-audit discussions, as represented in available conversation history.
- **Preparer:** ChatGPT (OpenAI GPT-6). Prepared at the Project Owner's request. The directive is a synthesis, not a claim of unanimous contributor approval.

Appendix C — Revision History

Version	Date	Prepared by	Change summary	Status
1.0	2026-11-08	ChatGPT (OpenAI GPT-6)	Initial publication-ready consolidation: project background, owner clarification, independent experimentation, failure provisions, evaluator audit, deliverables, governance, and reporting protocol.	Pre-Canon

1.0 / REGENERATED-02 | 2026-11-08 | ChatGPT (OpenAI GPT-6) | Second delivery-only regeneration after the user reported that both initial and first replacement PDF, DOCX, and Markdown files could not be downloaded. Original research content and semantic version remain v1.0; only regeneration metadata and tracking records added. | Pre-Canon; regenerated artifact |

Proposed next revision rule: Increment the patch version for editorial corrections, the minor version for revised investigative procedures or appendices, and the major version for substantive changes in governance or research methodology. Record each change here and retain earlier released artifacts in Git history.

Appendix D — Artifact Regeneration and Download Incident Log

This appendix records **delivery attempts**, distinct from **research-document revisions**. A download failure is not evidence of a problem with the directive's research content. The regeneration identifier is appended to all three filenames so published copies and delivery incidents remain traceable.

Attempt	Date	Formats	Reported issue	Response / verification	Research version
ORIGINAL	2026-11-08	PDF, DOCX, MD	The Project Owner reported that the initial files could not be downloaded.	Initial delivery unavailable to the user; prior validation did not establish user-side download success.	v1.0
REGENERATED-01	2026-11-08	PDF, DOCX, MD	First replacement attempted; the Project Owner reported these links also failed.	Local file integrity checks passed, but user-side download unsuccessful.	v1.0
REGENERATED-02	2026-11-08	PDF, DOCX, MD, ZIP	Second replacement after repeated delivery failure.	Rebuilt from original directive with delivery log; ZIP fallback provided. Download status to be confirmed by recipient.	v1.0

Future naming convention: OAL-CURR-RESEARCH-INVESTIGATION-001_v1.0_REGENERATED-03 for the next replacement set, if needed; continue -04, -05, etc. Change the v1.0 element only when the actual research directive is revised. Keep all previous filenames and incident descriptions in repository history.

Distinction: "REGENERATED" does not imply a new canonical research revision or successful end-user download. A technical build or validation pass confirms only that the generated files are structurally readable in the producing environment.

End of document — OAL-CURR-RESEARCH-INVESTIGATION-001 v1.0 / REGENERATED-02